

Geometry

Unit 12

Lesson 4 Homework

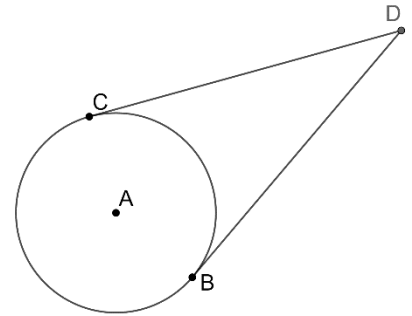
Name _____

Date _____

Period _____

For questions 1 through 4, Circle A is shown. $\overline{CD}$ and $\overline{DB}$ are tangent to the circle.

$$CD = 10x + 30 \text{ and } DB = 4x + 66.$$

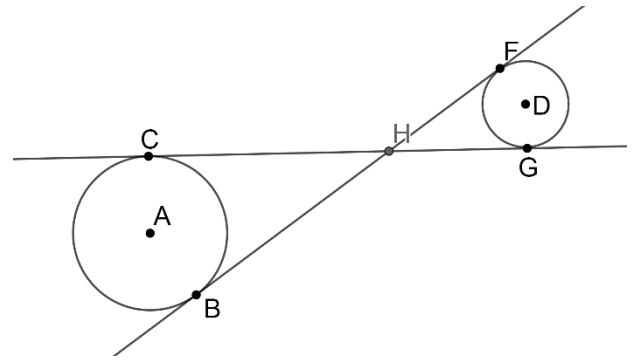


1. What is the value of x ?

2. What is the length of $\overline{CD}$?

3. What is the length of $\overline{DB}$?

For questions 4 through 10, Circles A and D are shown. Lines $\overline{CG}$ and $\overline{BF}$ intersect at point H . Line $\overline{CG}$ is tangent to circle A at point C and circle D at point G . Line $\overline{BF}$ is tangent to circle A at point B and circle D at point F . $CH = 3x - 8$, $HG = 5y$, $BH = 2x - 3$, $HF = 3y + 2$.



4. Write an equation you can use to solve for x .
5. Solve the equation you wrote in question 4 for x .
6. Write an equation you can use to solve for y .
7. Solve the equation you wrote in question 6 for y .
8. What is the length of $\overline{BF}$?
9. What is the length of $\overline{CH}$?
10. What is the length of $\overline{HF}$?